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Inconsistent convergence in Hilbert space

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Abstract

Another problem with Hilbert spaces in quantum mechanics is that unbounded operators are not continuous over their own domain, which leads to physical inconsistencies. We show this by creating a sequence of states for the harmonic oscillator that converges to the ground state while the expected energy level converges to the first one.

1 Introduction

Due to the completeness requirement, the domain of an unbounded self-adjoint operator cannot be the whole space. Similarly, any continuous linear operator must be bounded, since if it is defined on a dense subspace it could be extended to the whole space. This means that any unbounded operator not only is not defined over the whole space (e.g. the average position is not defined for all states), but it is not continuous in its own domain (i.e. small state changes can correspond to big average position changes). This means that states and observables can converge in different ways for the same sequence.

2 Sequence that converges in ℓ^2 but not in $S(\mathbb{N})$

Recall that ℓ^2 is the space of sequences x_i such that $\sum_{i=1}^{\infty} |x_i|^2 < \infty$. Since a probability distribution over countably many elements is a sequence p_i such that $p_i \in [0,1]$ and $\sum_{i=1}^{\infty} p_i = 1$, we have $p_i \in \ell^2$. Convergence in ℓ^2 coincides with convergence of the inner product. That is, given a sequence of sequences $\{x_i^j\}_{j=1}^{\infty}$, $\lim\limits_{j\to\infty} x_i^j \to y_i$ if and only if $\lim\limits_{j\to\infty} d(x_i^j, y_i) \to 0$ where $d(x_i^j, y_i) = |x_i^j - y_i|^2 = \sum_{i=1}^{\infty}(x_i^j - y_i)^2$.

The space of Schwartz sequences $S(\mathbb{N}) \subset \ell^2$ consists of all sequences such that $\sum_{i=1}^{\infty} |i^n x_i|^2 < \infty$ for all n.

Proposition 1. *Let a_i be such that $a_1 = 1$ and $a_i = 0$ for all $i > 1$. Let $\{b_i^j\}_{j=2}^{\infty}$ be the sequence of sequences such that $b_i^j = 1$ if $i = j$ and $b_i^j = 0$ if $i \neq j$. Let $e_i^j = \sqrt{\frac{j-1}{j}} a_i + \sqrt{\frac{1}{j}} b_i^j$. Then $\lim\limits_{j\to\infty} e_i^j = a_i$ in ℓ^2 while e_i^j does not converge in $S(\mathbb{N})$ since $\lim\limits_{j\to\infty} \sum_{i=1}^{\infty} |i e_i^j|^2 \to +\infty$.*

Proof. Intuitively, for a fixed j, $|b_i^j|^2$ is a probability distribution over the single element at j. Each probability distribution b_i^j is in $S(\mathbb{N})$ but the sequence has no limit in either $S(\mathbb{N})$ or ℓ^2 because the probability escapes at infinity.

The convex combination e_i^j, instead, converges to a_i in ℓ^2 as $j \to \infty$. In fact

$$\begin{aligned}
d(e_i^j, a_i) &= \sum_{i=1}^{\infty} (e_i^j - a_i)^2 = \sum_{i=1}^{\infty} \left(\sqrt{\frac{j-1}{j}} a_i + \sqrt{\frac{1}{j}} b_i^j - a_i \right)^2 \\
&= \sum_{i=1}^{1} \left(\sqrt{\frac{j-1}{j}} a_i - a_i \right)^2 + \sum_{i=2}^{\infty} \left(\sqrt{\frac{1}{j}} b_i^j \right)^2 \\
&= \left(\sqrt{\frac{j-1}{j}} - 1 \right)^2 a_1^2 + \frac{1}{j} (b_j^j)^2 = \left(\sqrt{\frac{j-1}{j}} - 1 \right)^2 + \frac{1}{j} \\
\lim_{j\to\infty} d(e_i^j, a_i) &= \lim_{j\to\infty} \left(\left(\sqrt{\frac{j-1}{j}} - 1 \right)^2 + \frac{1}{j} \right) = 0
\end{aligned} \tag{2}$$

We also have

$$\begin{aligned}
\sum_{i=1}^{\infty} |i\, e_i^j|^2 &= \sum_{i=1}^{\infty} i^2 \left(\sqrt{\frac{j-1}{j}} a_i + \sqrt{\frac{1}{j}} b_i^j \right)^2 \\
&= \sum_{i=1}^{\infty} i^2 \left(\sqrt{\frac{j-1}{j}} a_i \right)^2 + \sum_{i=1}^{\infty} i^2 \left(\sqrt{\frac{1}{j}} b_i^j \right)^2 \\
&= 1^2 \cdot \frac{j-1}{j} \cdot a_1^2 + j^2 \cdot \frac{1}{j} \cdot (b_j^j)^2 = \frac{j-1}{j} + j \\
\lim_{j\to\infty} \sum_{i=1}^{\infty} |i\, e_i^j|^2 &= \lim_{j\to\infty} \left(\frac{j-1}{j} + j \right) = +\infty
\end{aligned} \tag{3}$$

□

3 Inconsistent convergence for energy levels

Now, consider the harmonic oscillator with energy eigenstates $|n\rangle$ and number operator $N = a^\dagger a$ where $N|n\rangle = n|n\rangle$. Based on the previous discussion, the sequence of states $|e^j\rangle = \sqrt{\frac{j-1}{j}}|0\rangle + \sqrt{\frac{1}{j}}|j\rangle$ converges to the ground state $|0\rangle$ in ℓ^2. However, recalling that the cross terms vanish since N is diagonal in the energy eigenbasis, the expected energy level is

$$\langle e^j|N|e^j\rangle = \frac{j-1}{j}\langle 0|N|0\rangle + \frac{1}{j}\langle j|N|j\rangle = \frac{j-1}{j}0 + \frac{1}{j}j = 1 \tag{4}$$

for all j. That is,

$$\begin{aligned}
\lim_{j\to\infty} |e^j\rangle &= |0\rangle \\
\lim_{j\to\infty} \langle e^j|N|e^j\rangle &= 1
\end{aligned} \tag{5}$$

the state converges to the ground state but the expected energy level to the first excited level, which is obviously physically inconsistent. This is because ℓ^2 convergence only tracks the inner product, which is insensitive to how the state is distributed across high energy levels. The small but far component $\sqrt{\frac{1}{j}}|j\rangle$ vanishes in the ℓ^2 norm but carries enough energy to keep the expectation fixed. A topology that keeps track of all moments, like the Schwartz topology, would not allow this kind of convergence.

4 Conclusion

As we have shown, we created a sequence of quantum states for the harmonic oscillator that, in the standard framework of Hilbert spaces, converges to the ground state while the expectation for the energy level converges to the first level. This is clearly physically inconsistent, and shows how and why Hilbert spaces are not suitable as a physically meaningful foundation for quantum mechanics. The problem is that the topology is too weak to take into account all conditions for a meaningful criterion of physical convergence.